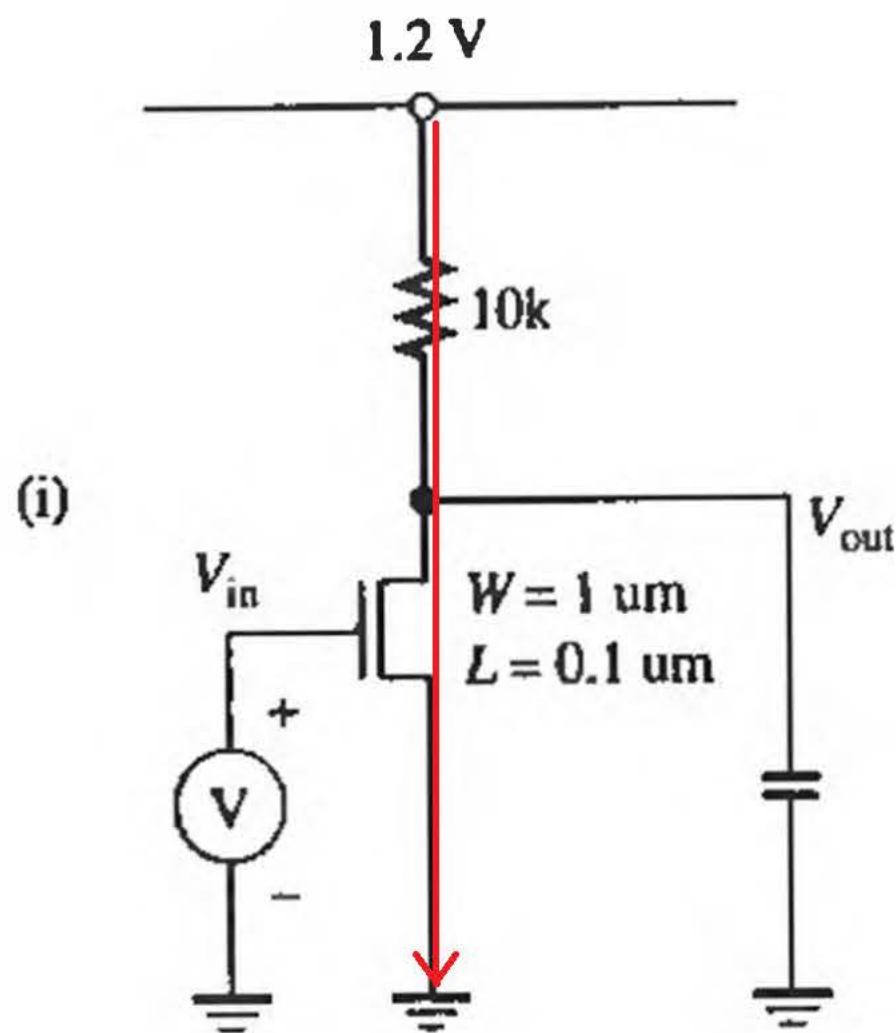
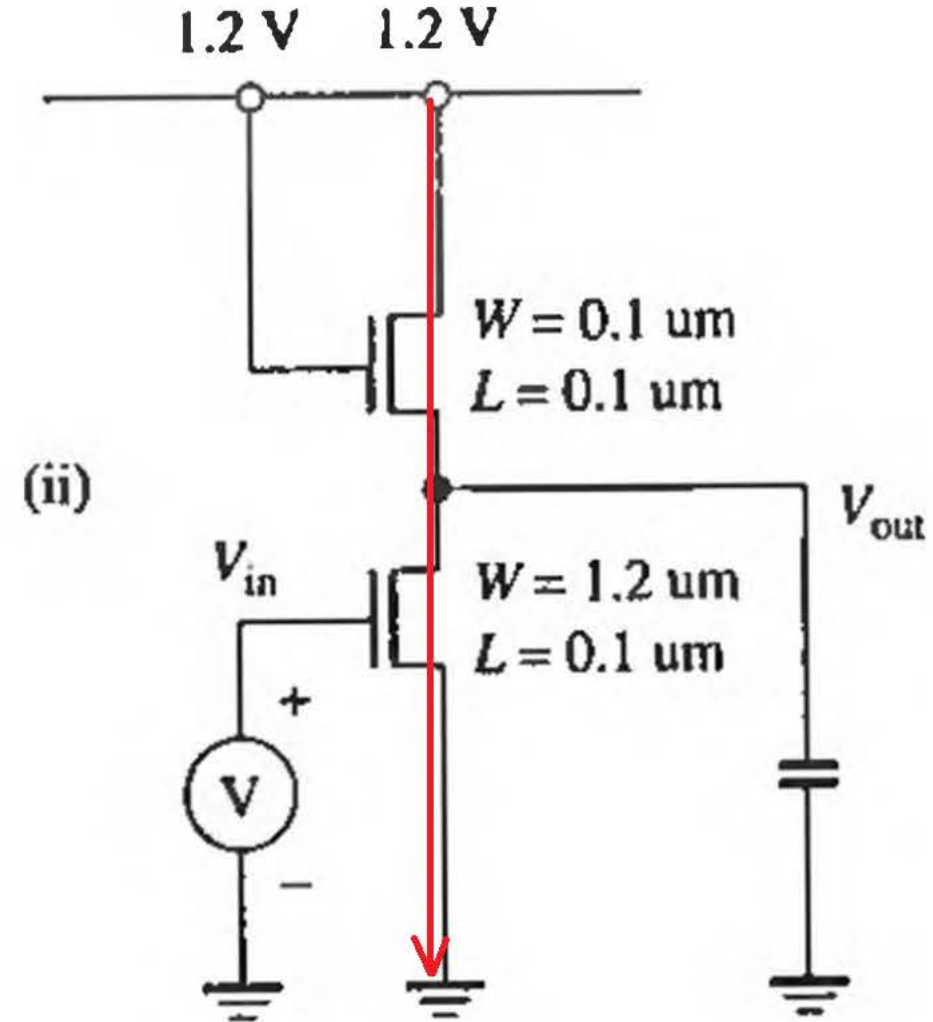


1.a

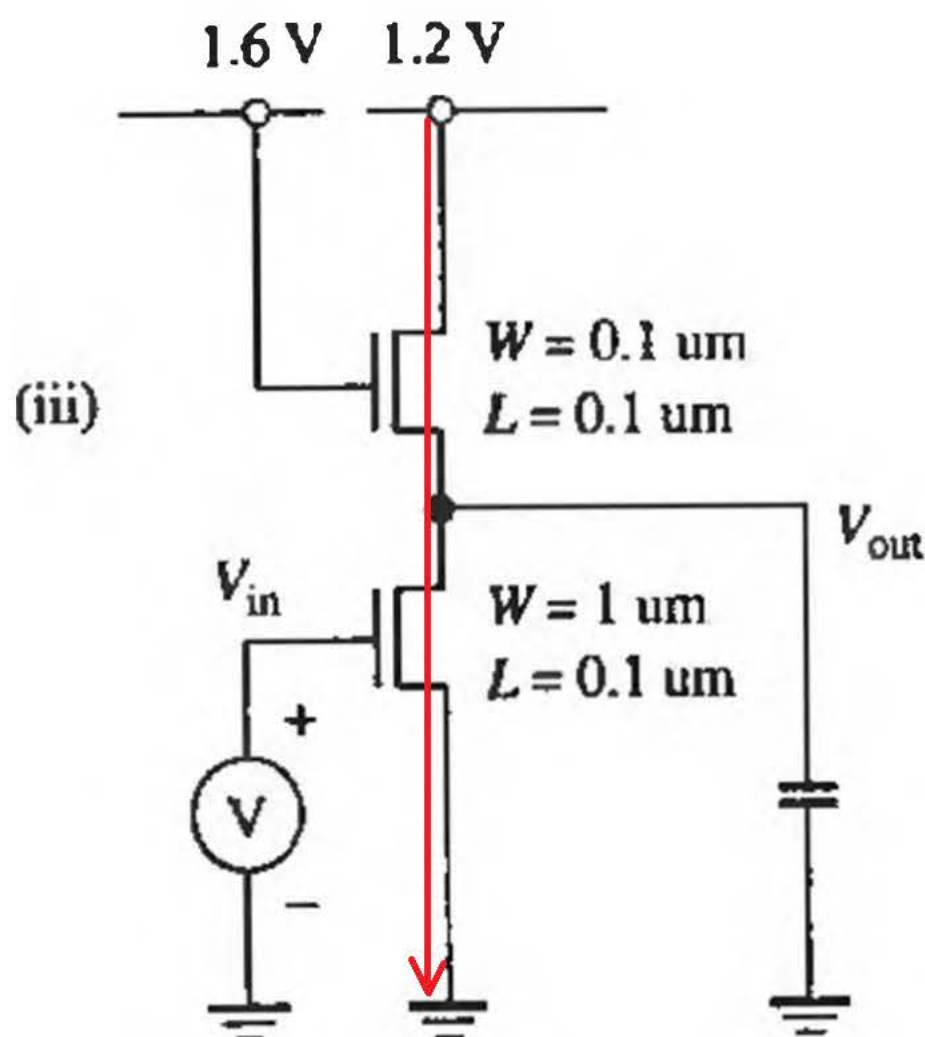
$V_{in} = \text{logic high}$



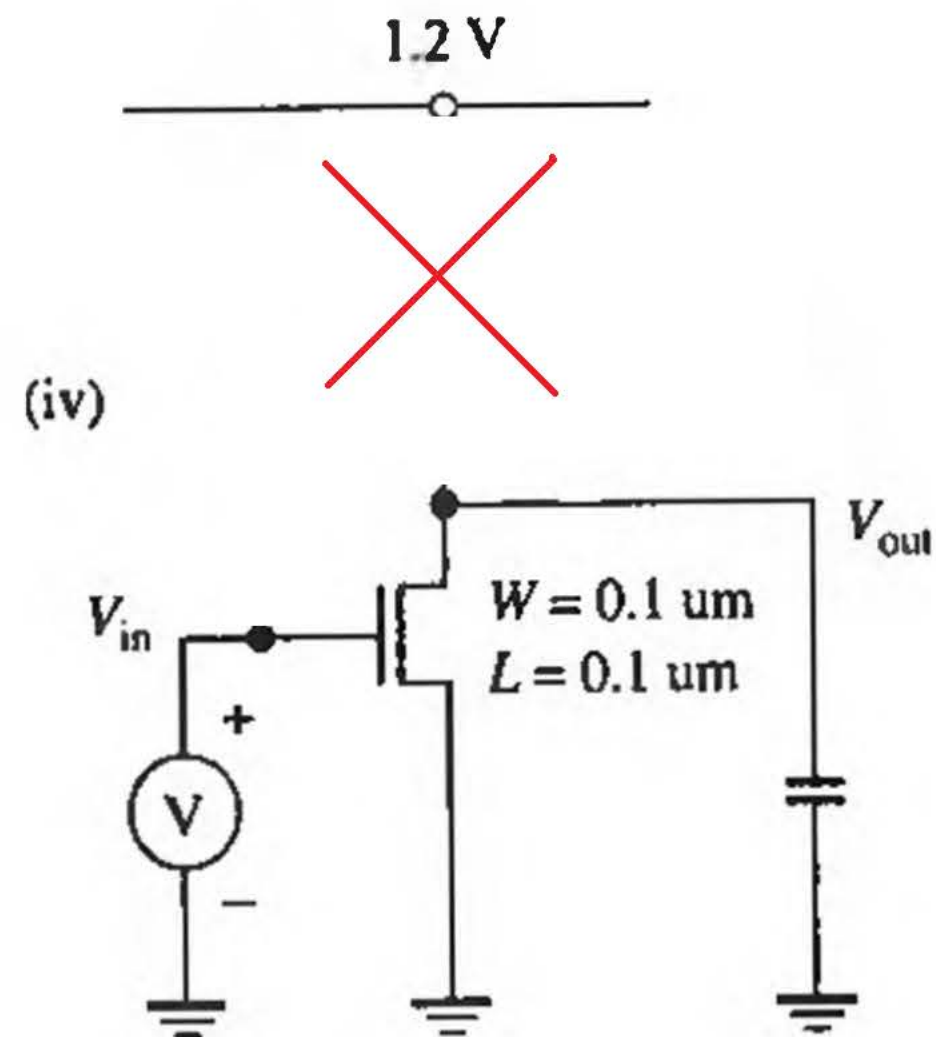
Resistor always conducts current with any non-zero voltage differential across it. Constant current flow from VDD to GND.



Both NMOS are on. Constant current flow from VDD to GND.



Both NMOS are on. Constant current flow from VDD to GND.

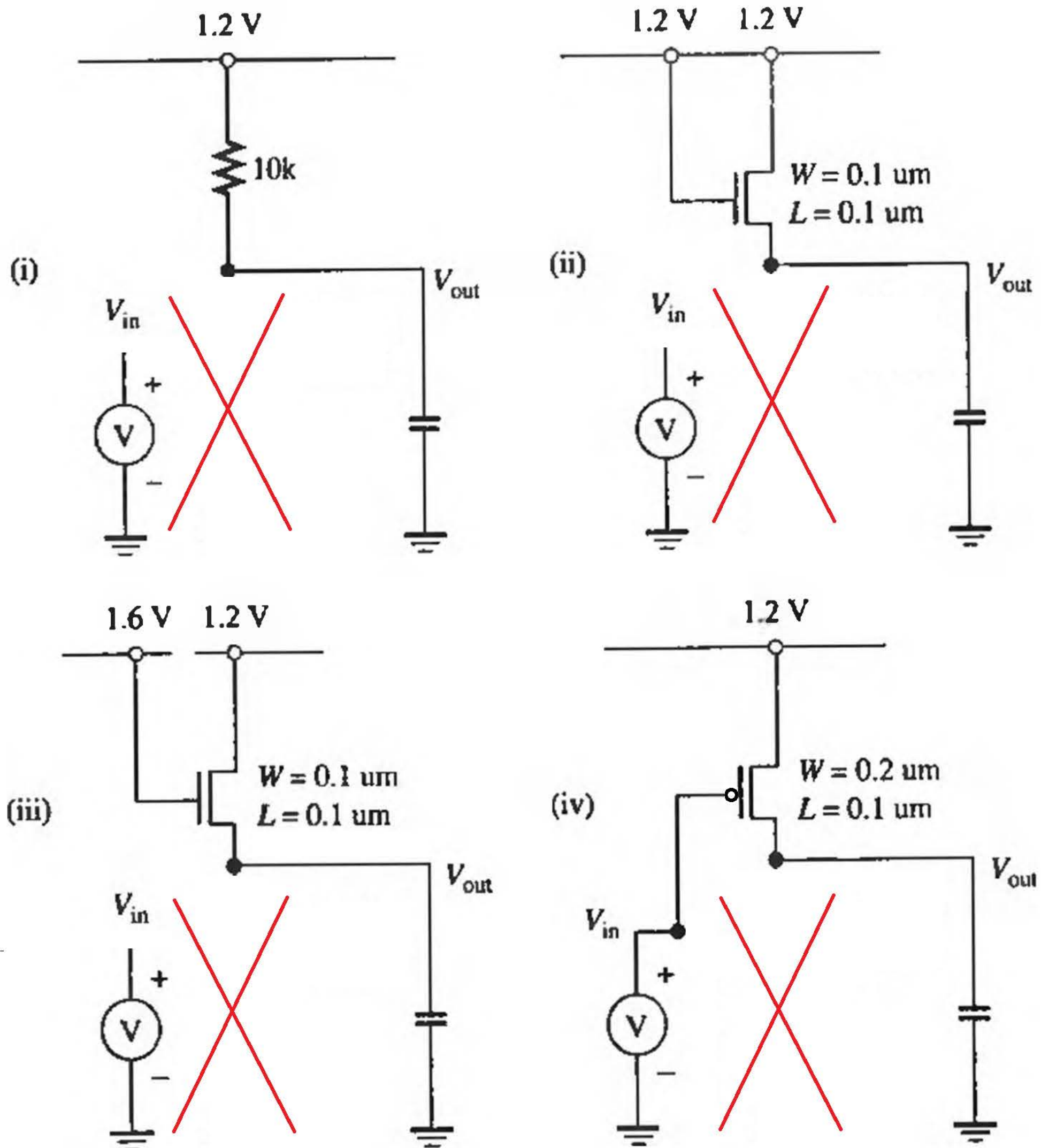


PMOS pull-up is off. No connection to VDD. No steady-state current flow.

Cases i, ii, and iii consume static power when V_{in} is logic high due to constant current flow from VDD to GND.

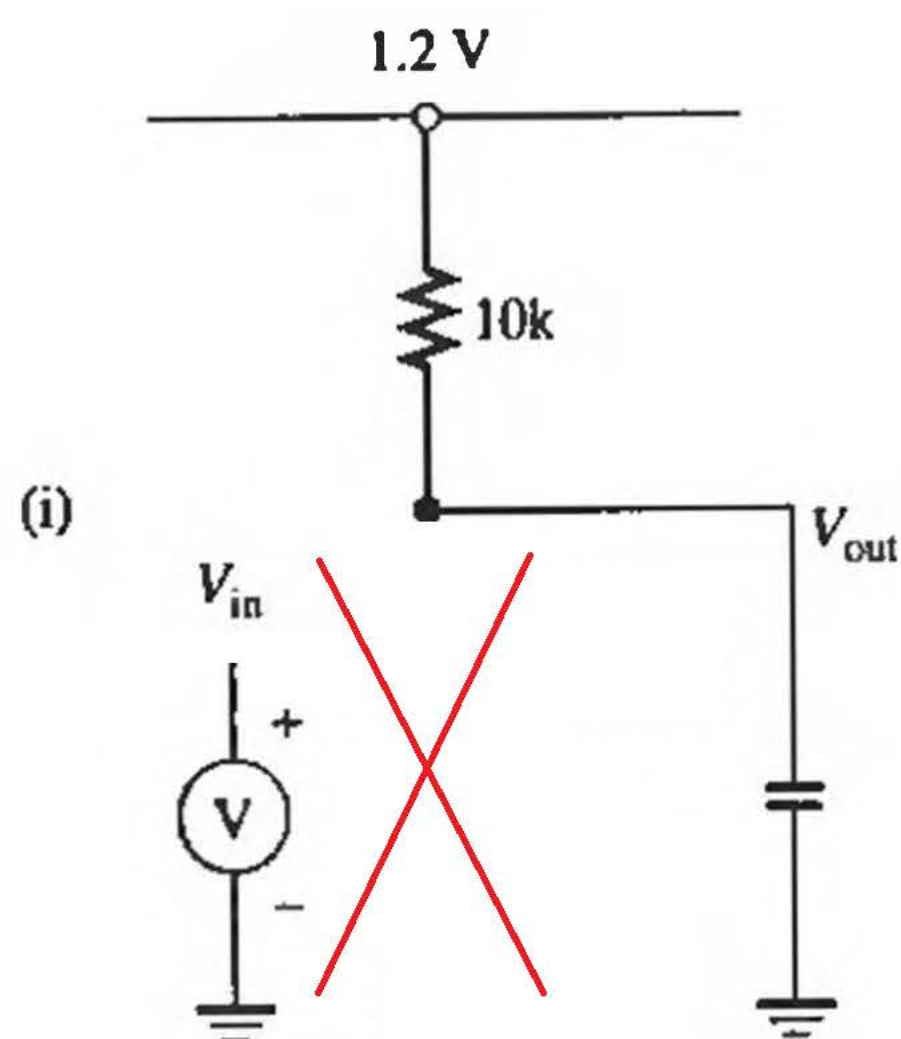
1.b

V_{in} Logic Low

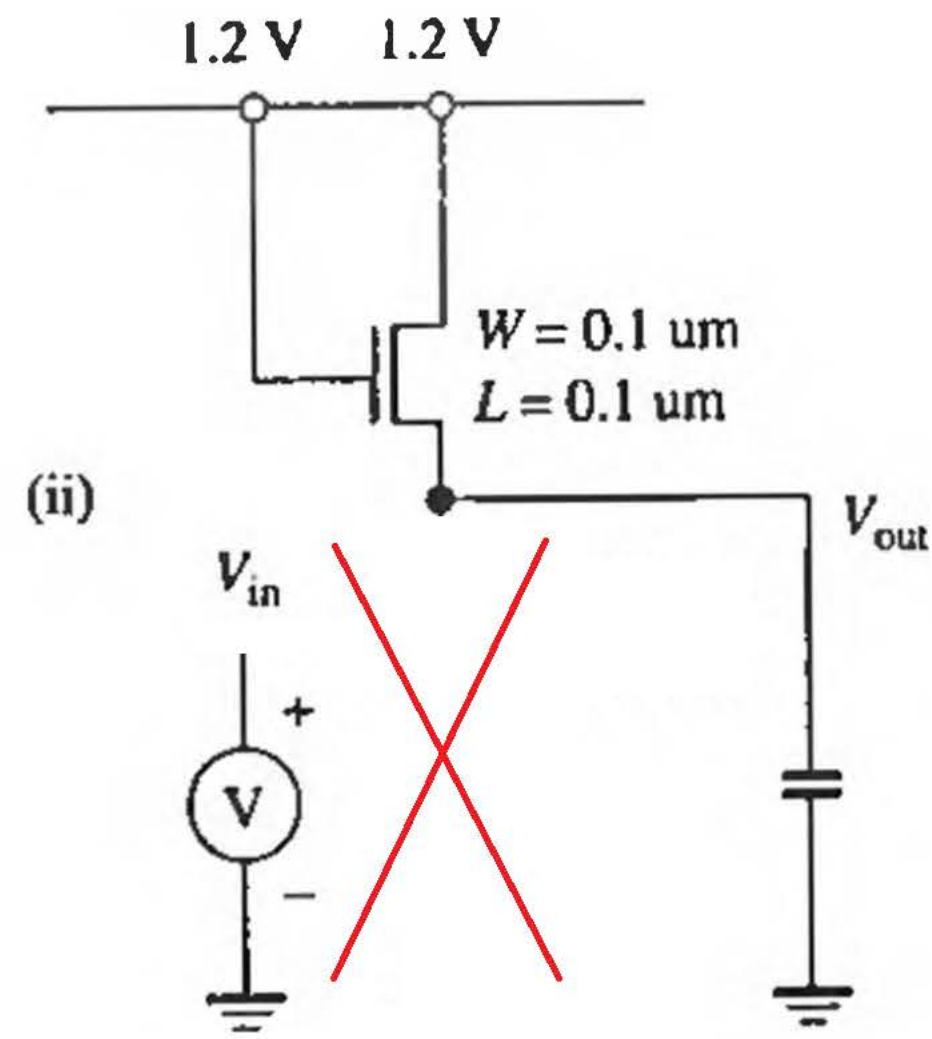


In all four cases i, ii, iii, and iv, the pull-down NMOS is off providing no current path between VDD and GND. All four cases do not have static power consumption when V_{in} is logic low.

$V_{in} = 0$ when VOH is high



VOH is 1.2V. Resistor will conduct as long as V_{out} is less than 1.2V to pull all the way to 1.2V. NMOS is off and will not contest with the resistor.



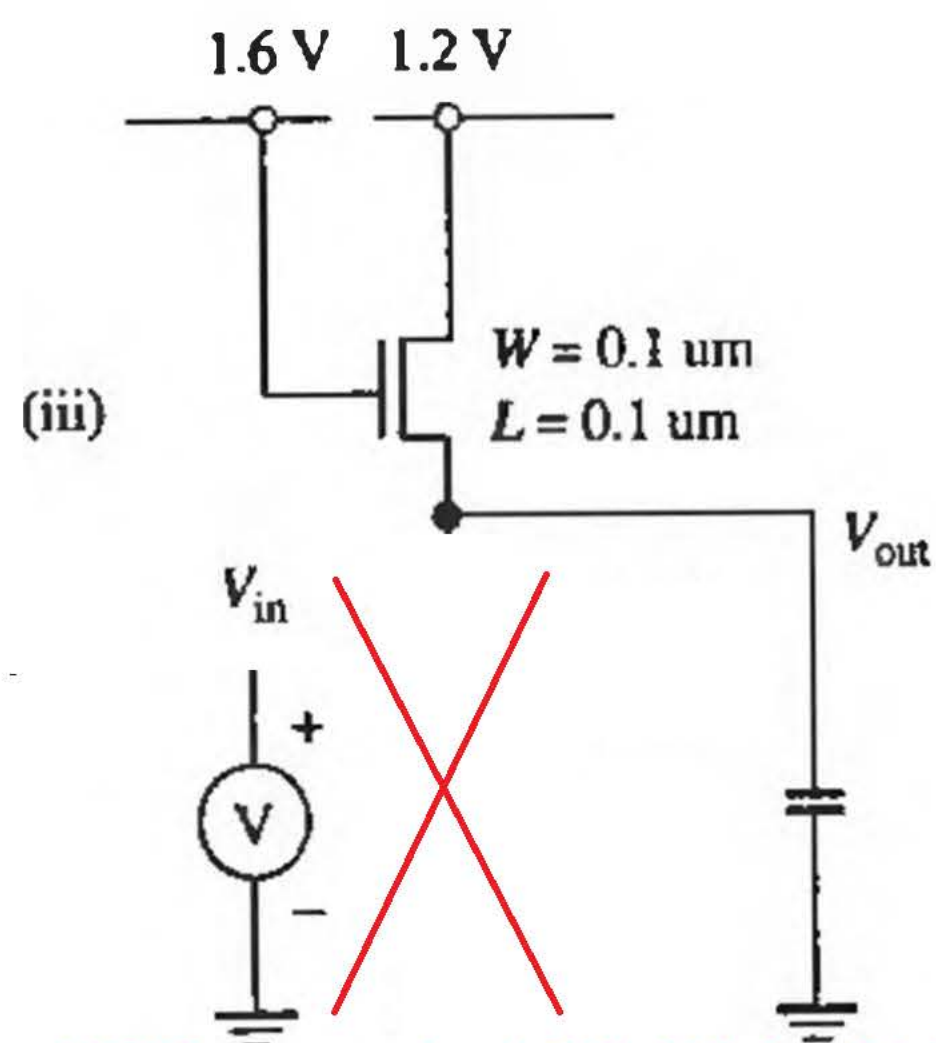
VOH cannot be 1.2V due to $V_{gs} - V_t$ limitation with an NMOS functioning as a pull-up. With body effect ignored, best case VOH is 0.8V.

$$V_{gs} - V_t = 0$$

$$V_g - V_s - V_t = 0$$

$$\text{If } V_t = V_{t0}, 1.2 - V_s - 0.4 = 0$$

$$V_s = V_{out} = 0.8V$$



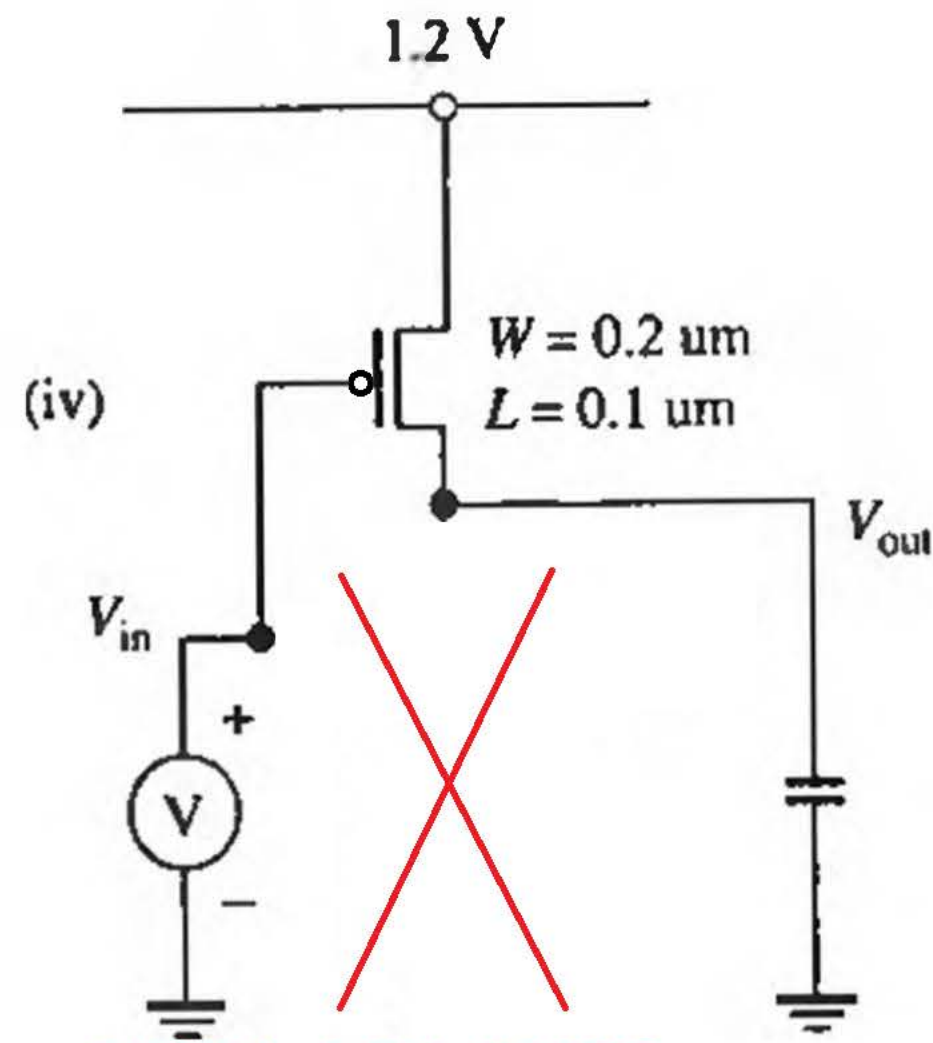
VOH cannot be 1.2V. With a V_{t0} of 0.4V the pull-up NMOS could potentially reach 1.2V, but source will be at higher voltage than grounded bulk for a positive V_{sb} term, increasing V_t above 0.4V.

$$V_{gs} - V_t = 0$$

$$V_g - V_s - V_t = 0$$

$$\text{If } V_t = V_{t0}, 1.6 - V_s - 0.4 = 0$$

$$V_s = V_{out} = 1.2V, \text{ but } V_t > V_{t0}$$

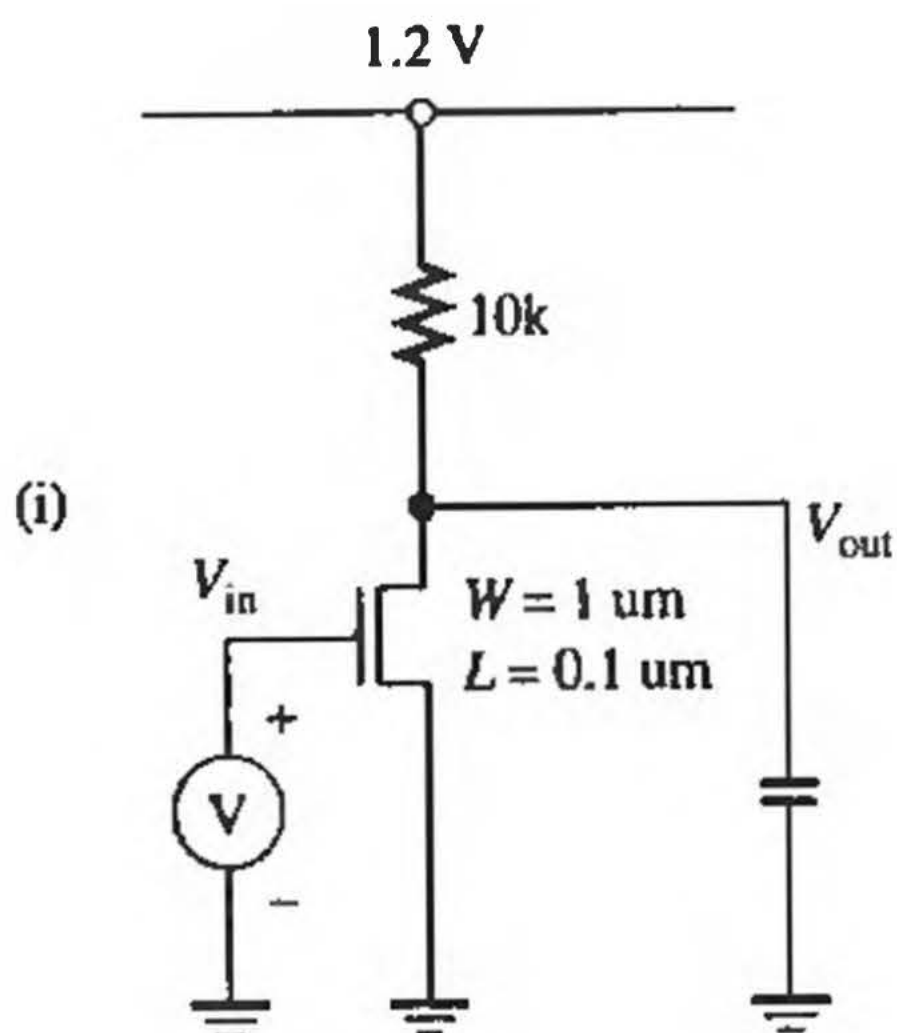


VOH is 1.2V. PMOS as pull-up does not have a $V_{sg} + V_t > 0$ limitation as Source is fixed to V_{dd} . Pull-down NMOS is off and will not contest with PMOS.

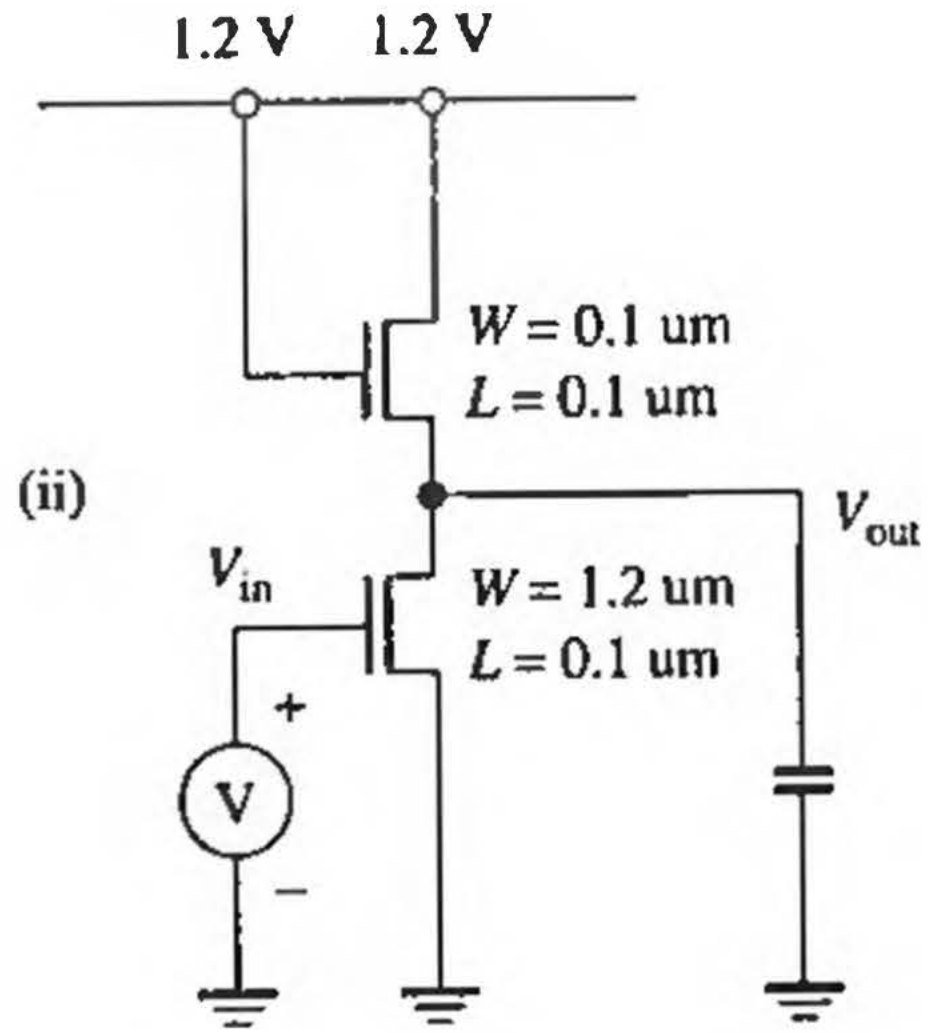
Cases i and iv have a VOH of 1.2V.

1.d

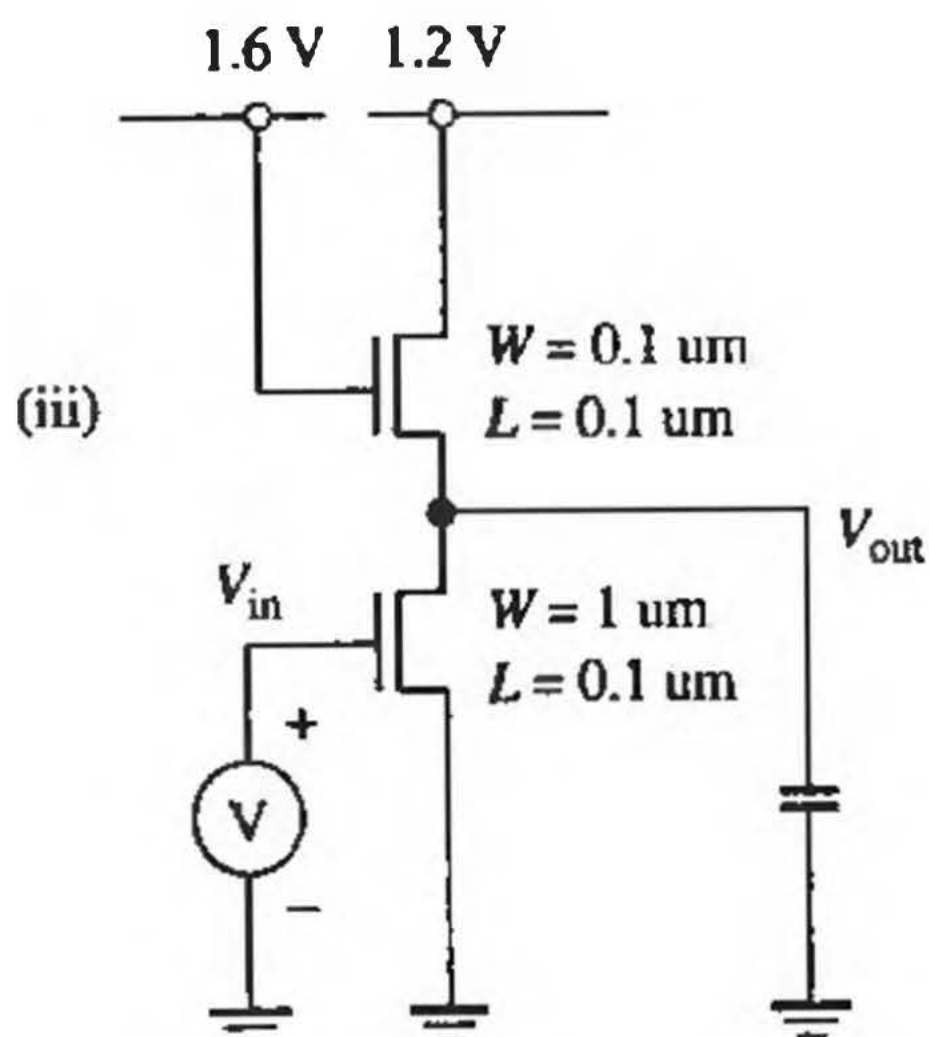
For $V_{OL} = \text{logic low}$, V_{in} is logic high



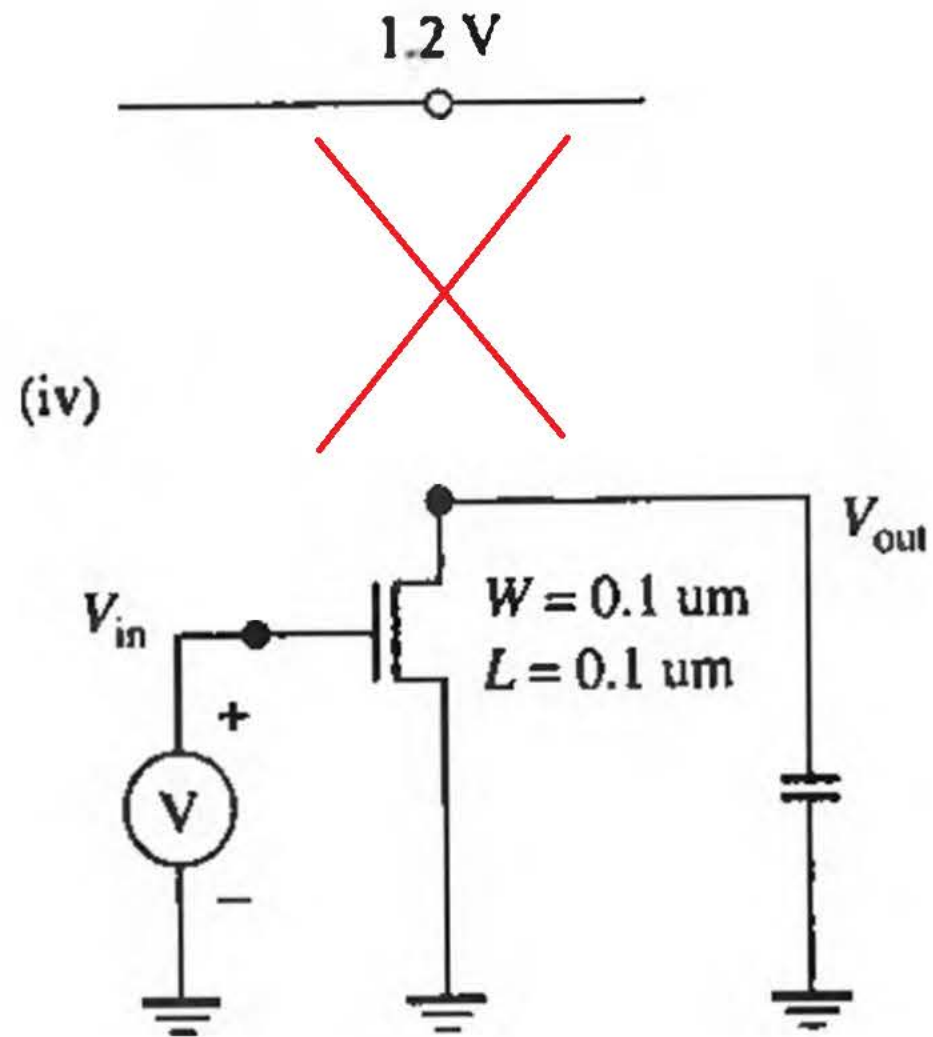
Pull-up resistor can never turn off and will contest the NMOS pull-down for V_{out} low, preventing V_{OL} from reaching 0.



Pull-up NMOS can never turn off and will contest the NMOS pull-down for V_{out} low, preventing V_{OL} from reaching 0

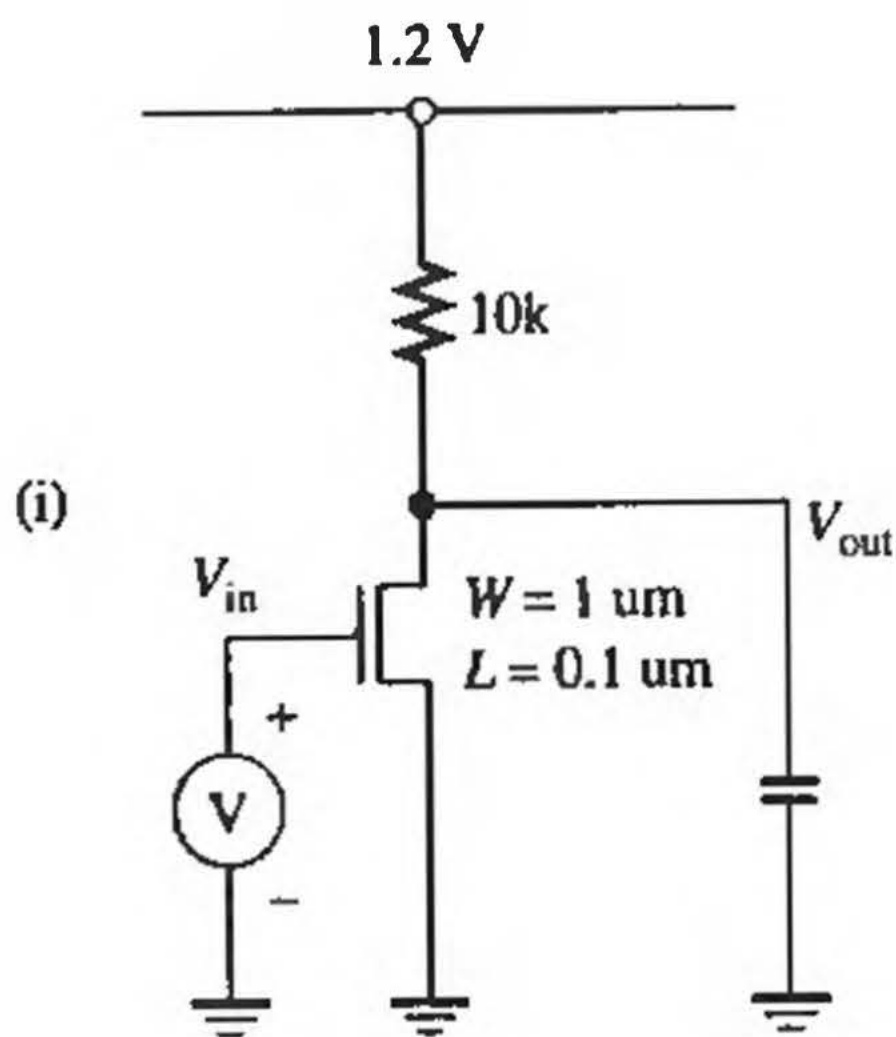


Pull-up NMOS can never turn off and will contest the NMOS pull-down for V_{out} low, preventing V_{OL} from reaching 0

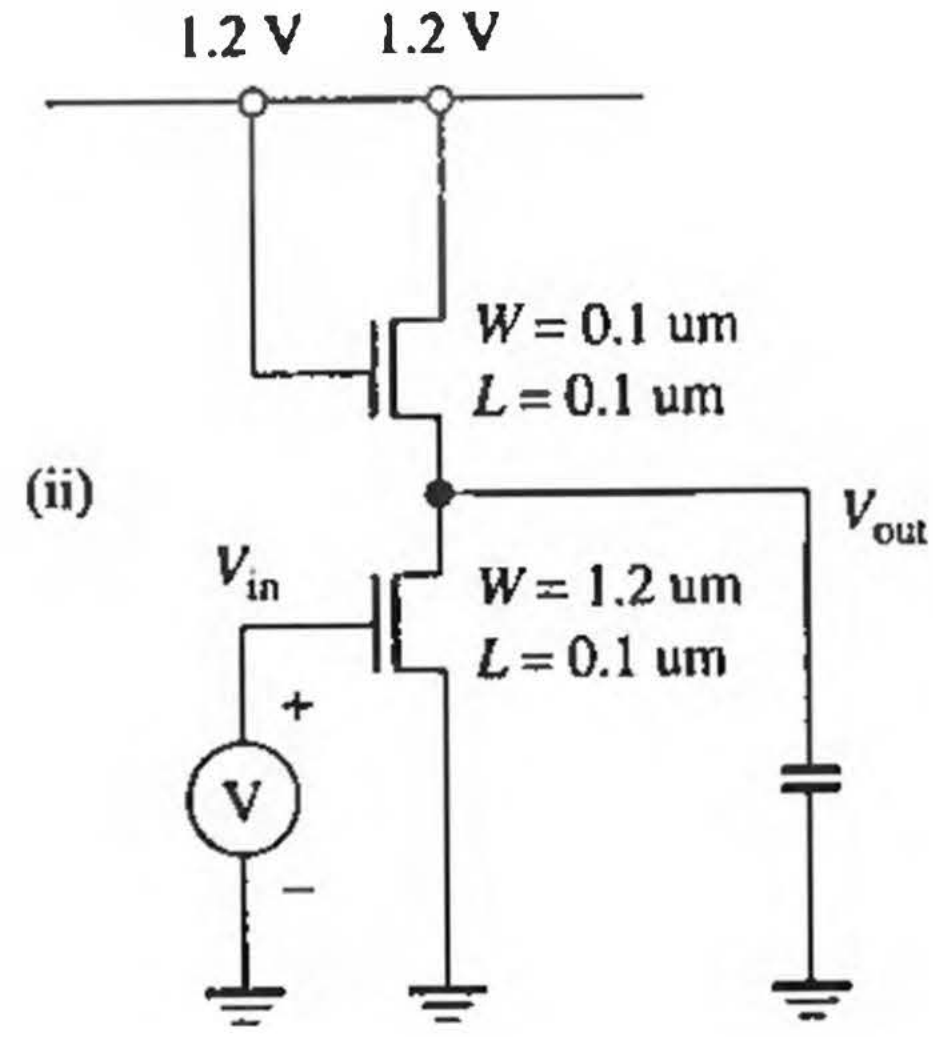


Pull-up PMOS will be off due to V_{in} being logic high. The pull-down NMOS will not be contested and can pull V_{out} down to 0 for a V_{OH} of 0V.

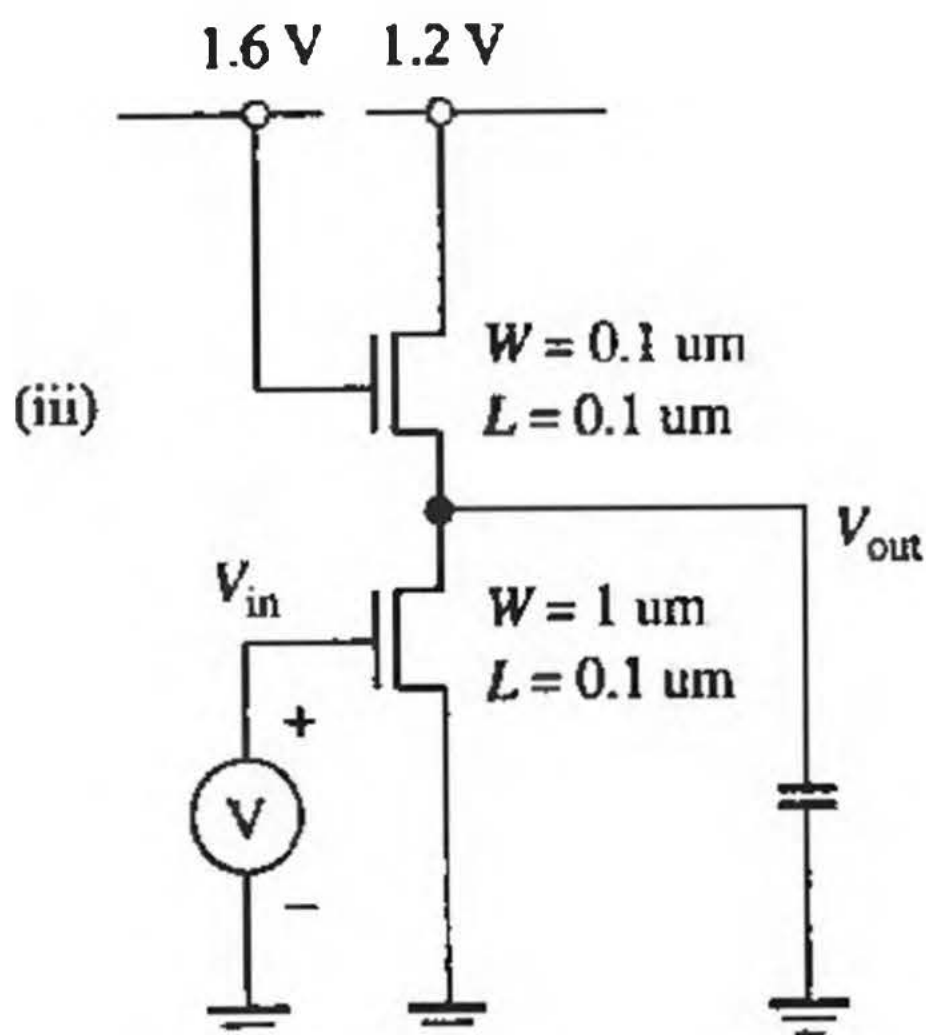
Only case iv has a V_{OL} of 0V.



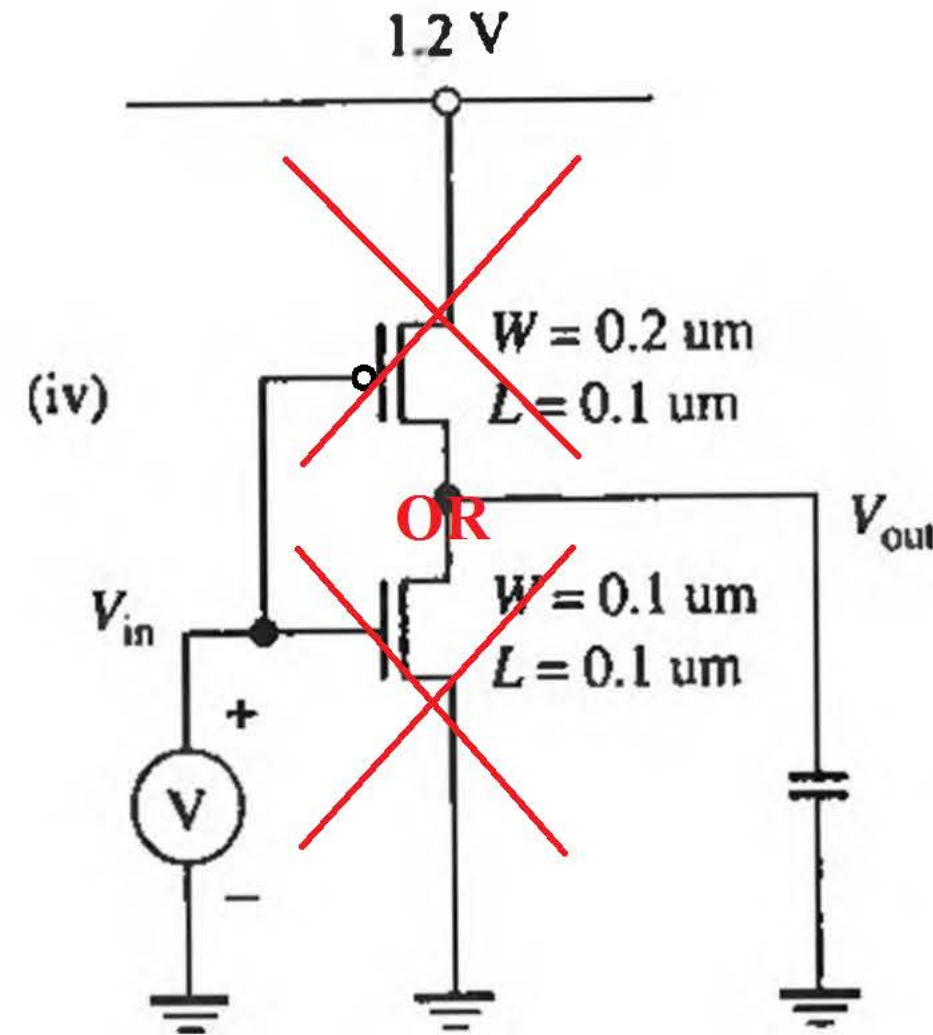
If W is too low, the NMOS on-state will not have a resistance that can overpower the always-on resistor.



If Pull-up W is too large or Pull-down W is too small, the pull-down NMOS cannot overpower the always-on pull-up NMOS for the output low scenario.



If Pull-up W is too large or Pull-down W is too small, the pull-down NMOS cannot overpower the always-on pull-up NMOS for the output low scenario.



In the output high scenario the pull-down NMOS is fully off and cannot contest the pull-up PMOS. In the output low scenario the pull-up PMOS is fully off and cannot contest the pull-down NMOS. There is no width constraint for proper operation as there is no output contention.

2]

$$a) \boxed{V_{OH} = 1.2V}$$

For V_{OL} ,

$$V_{out} = V_{DS}$$

$$I_R = I_{DS} = \frac{\mu_n C_{ox}}{1 + \frac{V_{DS}}{E_c L}} \cdot \frac{W}{L} \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] (1 + \chi V_{DS})$$

$$\frac{1.2 - V_{out}}{10k} = \frac{270 \cdot 1.6 \cdot 10^{-6} \cdot 10 \left[(1.2 - 0.4) V_{out} - \frac{V_{out}^2}{2} \right] (1 + 0.2 V_{out})}{1 + \frac{V_{out}}{0.6}}$$

No V_{SB} so $V_T = V_{T0}$

For full short channel solution, use solver to aid, such as Matlab, or WolframAlpha.

$$V_{out} = 0.0363V$$

Linear check

$$V_{DS} \leq V_{GS} - V_T$$

$$0.036 \leq 0.8$$

Linear confirmed

$$\boxed{V_{OL} = 0.0363V}$$

If doing low V_{OL} approximation:

$$V_{OL} \approx \frac{V_{DD}}{1 + \mu_n C_{ox} \frac{W}{L} (V_{DD} - V_T) R}$$

$E_c L$, χ have negligible impact when V_{DS} is very low

$$V_{OL} \approx \frac{1.2}{1 + 270 \cdot 1.6 \cdot 10^{-6} \cdot 10 (1.2 - 0.4) \cdot 10k}$$

$$V_{OL} \approx 0.0337V$$

ii) V_{OH}

onset of cutoff

$$V_{GS} - V_T = 0 \quad V_S = V_{out}$$

$$V_G - V_S - V_T = 0 \quad V_G = V_{DD}$$

$$1.2 - V_{out} - V_T = 0$$

V_T is not V_{T0} due to large V_{SB}

$$V_{SB} = V_S - V_B$$

$$V_{SB} = V_{out} - 0$$

$$V_{SB} = V_{out}$$

$$V_T = V_{T0} + \gamma (\sqrt{V_{SB} + 2\phi_F} - \sqrt{2\phi_F})$$

$$V_T = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$1.2 - V_{out} = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$V_{out} = 0.734V$$

$$\boxed{V_{OH} = 0.734V}$$

For V_{OL}

Assuming V_{OL} is low, upper NMOS will be saturated. Lower will be linear

$$I_{D_{upper}} = I_{D_{lower}}$$

assuming V_{OL} is low also assumes V_{DS} is low

$$W_u \mu_{ox} C_{ox} \frac{(V_{DD} - V_{out} - V_T)^2}{V_{DD} - V_{out} - V_T + E_{CL}} (1 + \lambda(V_{DD} - V_{out})) = \frac{\mu_n C_{ox}}{1 + \frac{V_{out}}{E_{CL}}} \cdot \frac{W_L}{L} \left[(V_{in} - V_T)V_{out} - \frac{V_{out}^2}{2} \right] (1 + \lambda V_{out})$$
$$10^{-5} \cdot 8 \cdot 10^6 \cdot \frac{(1.2 - V_{out} - 0.4)^2}{1.2 - V_{out} - 0.4 + 0.6} (1 + 0.2(1.2 - V_{out})) = \frac{2 \cdot 70 \cdot 12}{1 + \frac{V_{out}}{0.6}} \left[(1.2 - 0.4)V_{out} - \frac{V_{out}^2}{2} \right] (1 + 0.2 \cdot V_{out})$$

$$V_{out} = 0.0175 \text{ V}$$

For upper NMOS

$$V_{DS} \geq V_{GS} - V_T$$

$$V_{DD} - V_{out} \geq V_G - V_{out} - V_T$$

$$V_{DD} \geq V_G - V_T$$

$$1.2 \geq 1.2 - 0.4$$

$$1.2 \geq 0.8$$

saturation confirmed

For lower NMOS

$$V_{DS} \leq V_{GS} - V_T$$

$$V_{out} \leq V_{in} - V_T$$

$$0.0175 \leq 1.2 - 0.4$$

$$0.0175 \leq 0.8$$

linear confirmed

$$V_{OL} = 0.0175 \text{ V}$$

Due to low V_{OL} assumption, could have ignored lower NMOS short channel effects due to very low V_{DS} .

iii) V_{OH}

Work identical to part ii

$V_{GS} - V_T = 0$ at onset of cutoff

$$V_G - V_S - V_T = 0$$

$$1.6 - V_{out} - V_T = 0$$

$$V_{SB} = V_S = V_{out}$$

$$1.6 - V_{out} = V_T, \quad V_T = V_{T0} + \gamma (\sqrt{V_{out} + 2\phi_F} - \sqrt{2\phi_F})$$

$$V_T = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$1.6 - V_{out} = 0.4 + 0.2 (\sqrt{V_{out} + 0.88} - \sqrt{0.88})$$

$$V_{out} = 1.106 \text{ V}$$

$$V_{OH} = 1.106 \text{ V}$$

iii

Work is identical to that of ii, except V_g for upper NMOS is now 1.6 V, and W for lower NMOS is $1\mu\text{m}$.

Can do full short channel equations like in part ii solution, but will demonstrate here ignoring lower NMOS short channel effects due to very low V_{DS} .

$$I_{D\text{upper}} = I_{D\text{lower}}$$

$$W_u V_{sat} C_{ox} \frac{(1.6 - V_{out} - V_T)^2}{1.6 - V_{out} - V_T - E_{cl}} (1 + \lambda(V_{DD} - V_{out})) = \mu_n C_{ox} \frac{W_L}{L} \left[(V_{in} - V_T) V_{out} - \frac{V_{out}^2}{2} \right]$$

$$10^{-5} \cdot 8 \cdot 10^6 \frac{(1.6 - V_{out} - 0.4)^2}{1.6 - V_{out} - 0.4 + 0.6} (1 + 0.2(1.2 - V_{out})) = 270 \cdot 10 \cdot \left[(1.2 - 0.4) V_{out} - \frac{V_{out}^2}{2} \right]$$

$$\boxed{V_{OL} = 0.0359}$$

$V_{OL} = 0.0378$ using full short channel equation for lower NMOS

Note for upper NMOS

$$V_{DS} \geq V_{GS} - V_T$$

$$V_{DD} - V_{out} \geq V_G - V_S - V_T$$

$$V_{DD} - V_{out} \geq V_G - V_{out} - V_T$$

$$1.2 \geq 1.6 - 0.4$$

$$1.2 \geq 1.2$$

Upper NMOS is just at onset of saturation

Lower NMOS is linear with work identical to part ii

iv

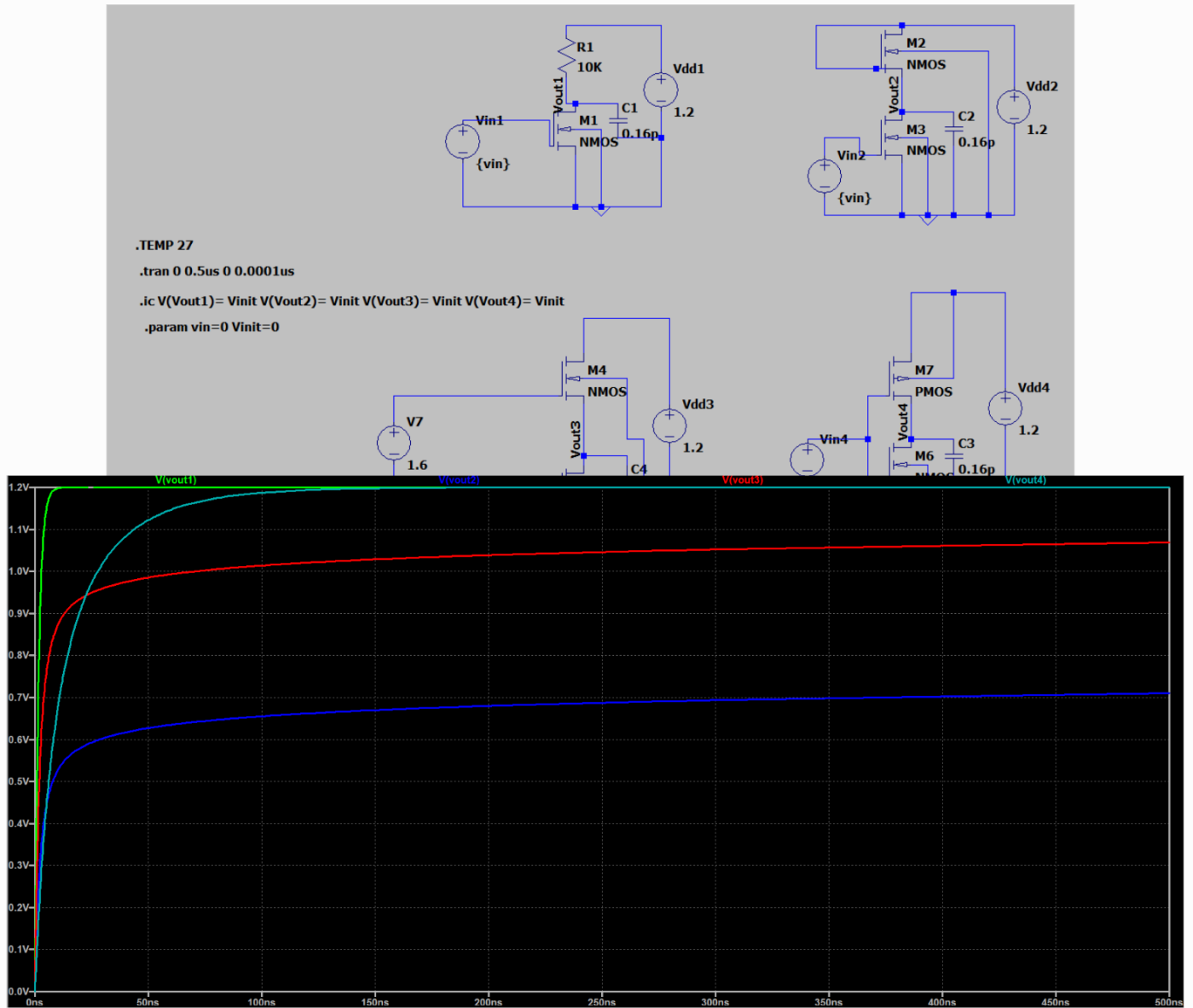
Since PMOS is off in output low state:

$$V_{OL} = 0V$$

Since NMOS is off in output high state:

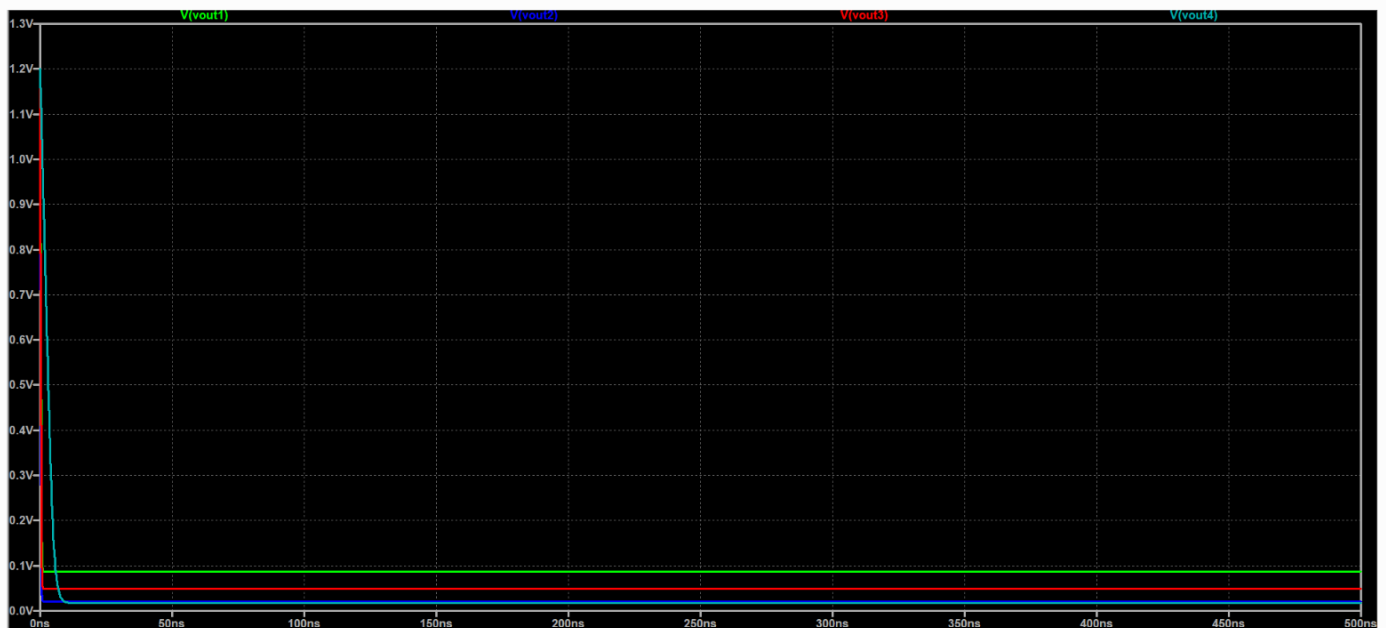
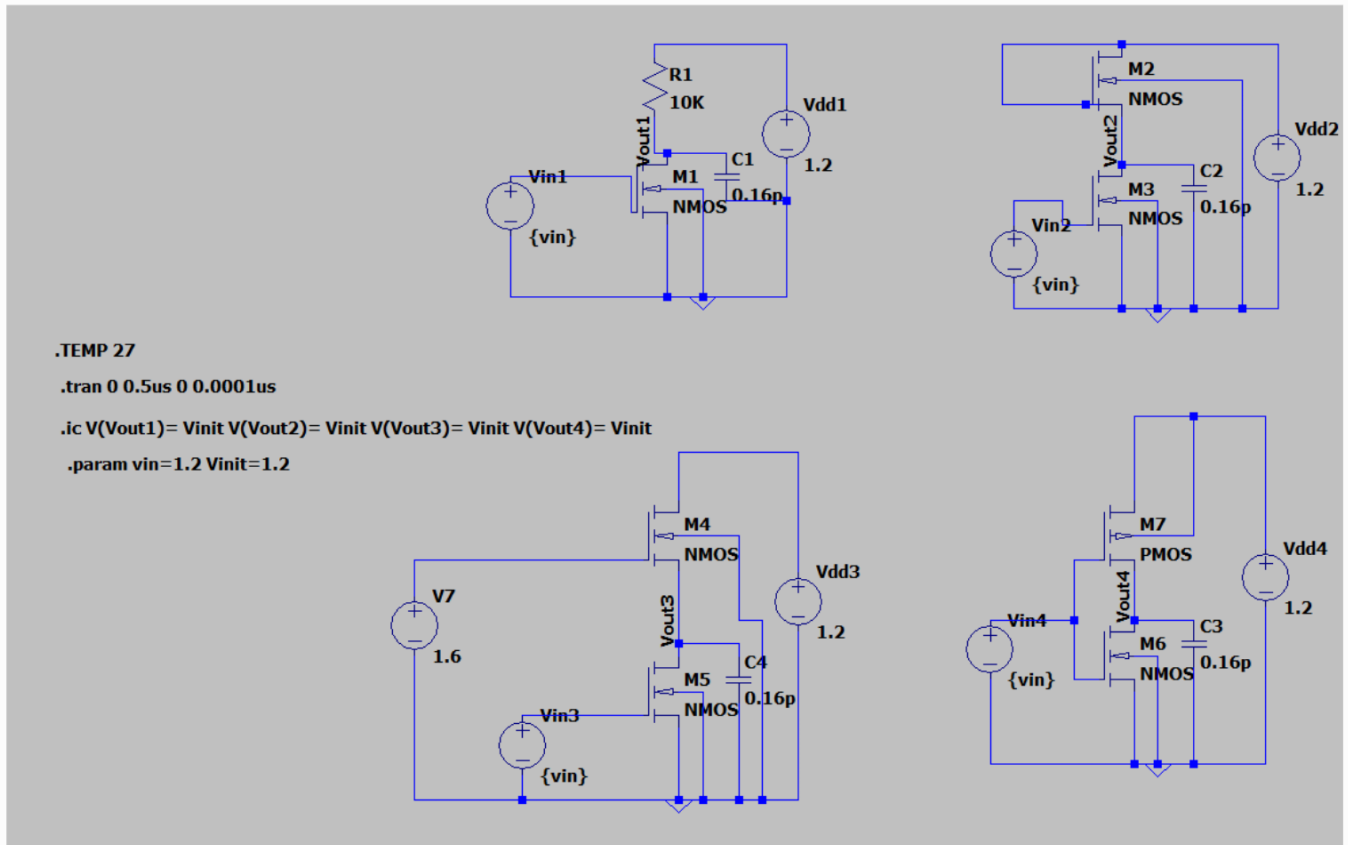
$$V_{OH} = 1.2V$$

V_{OH} :



Time scale is chosen in such a way that the transition is visible and the V_{OH} attains a stable value

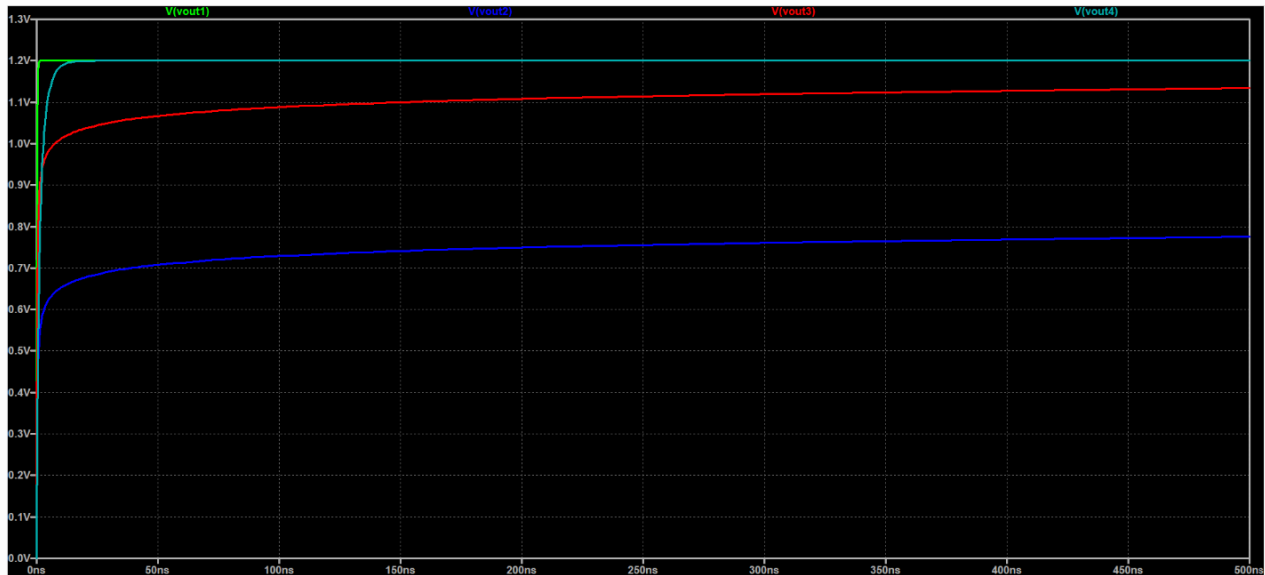
V_{OL} :



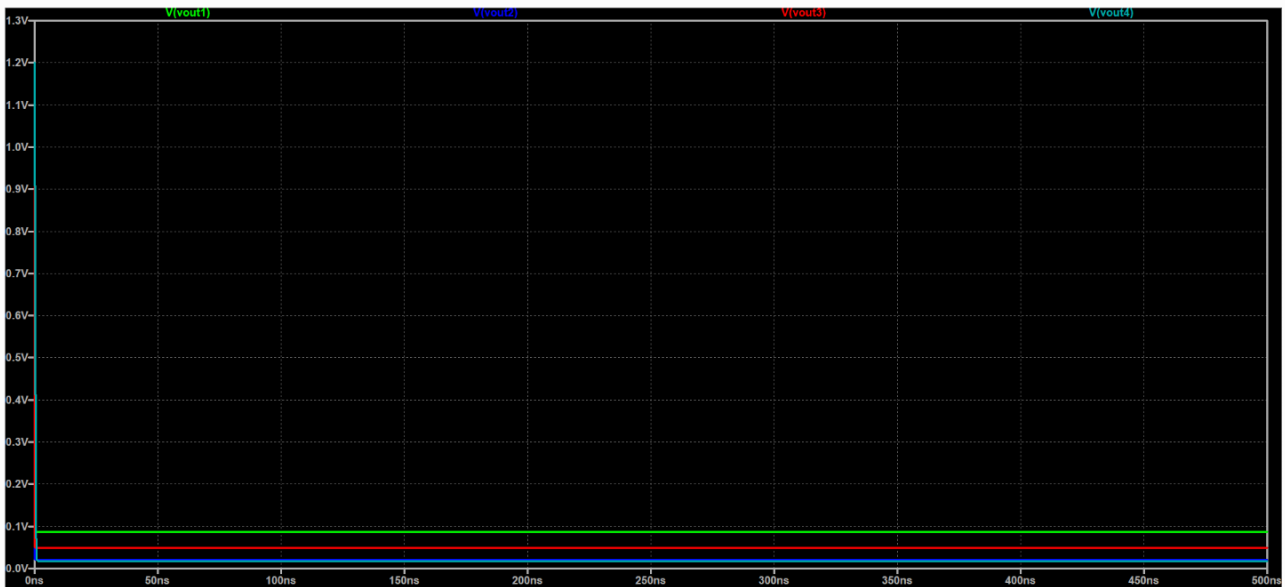
Both V_{OH} and V_{OL} values agree with our hand-calculated values.

→ Changing the capacitor to 0.016 pF gives us the following

V_{OH} :



V_{OL} :



Using the smaller capacitor allows us to reach V_{OL} and V_{OH} much rapidly. The difference in transition time is much more noticeable for CMOS compared to the other inverters. This affects the device performance greatly and known as delay which we will study in detail later in the course. Generally we want faster device, so a smaller capacitor is preferable.